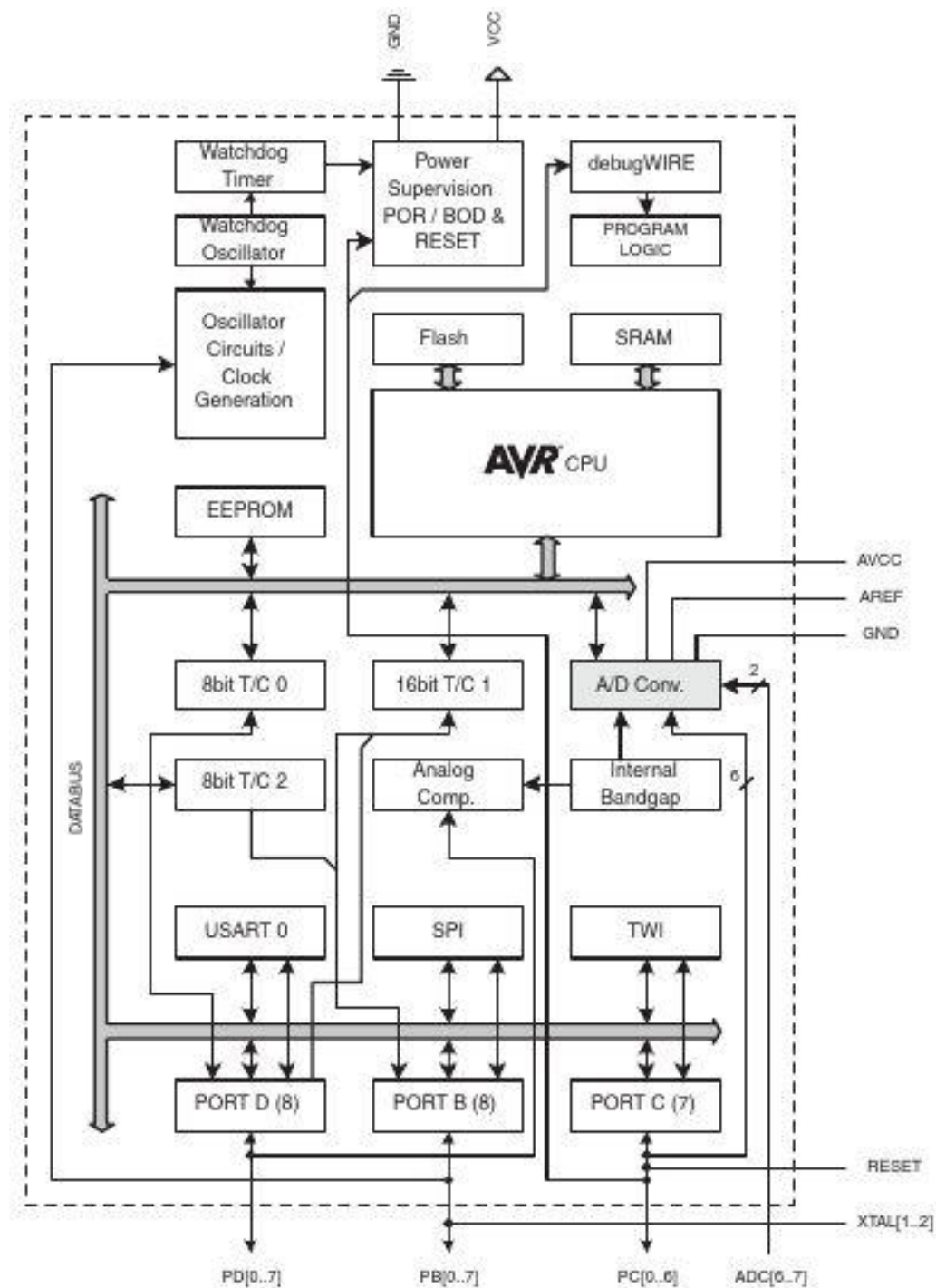
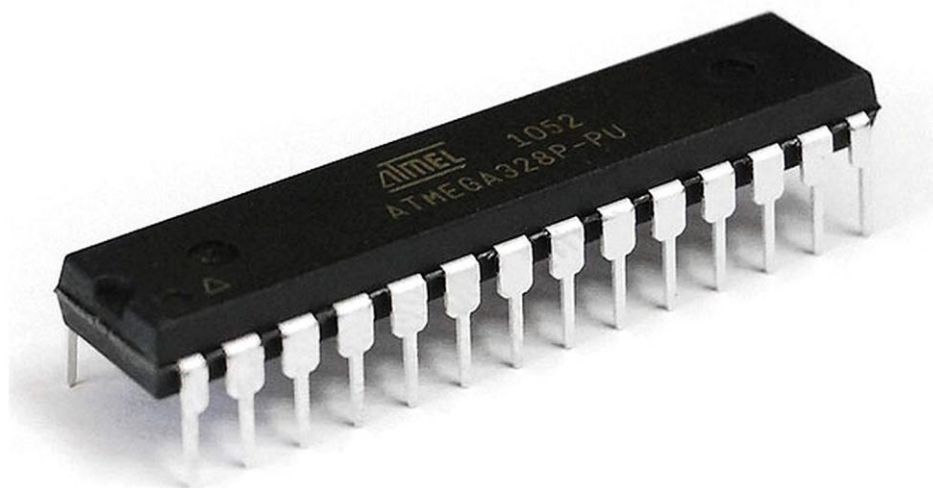


Arhitektura računalā

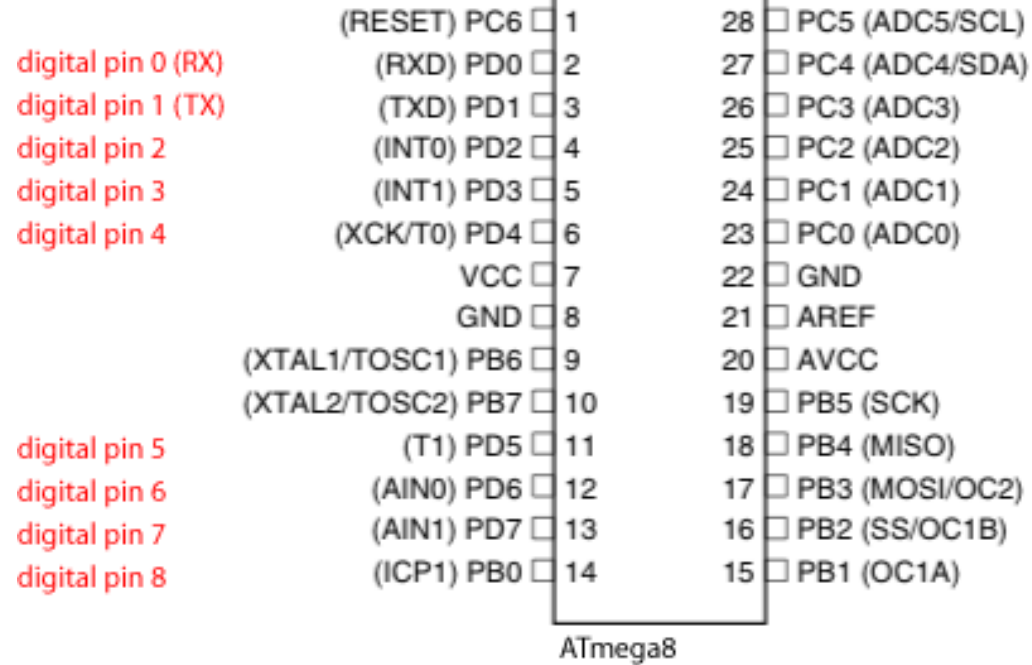
LV 7. Upravljanje portovima
I/O

Mikroupravljač ATMEGA 328p



Mikroupravljač ATMEGA 328p

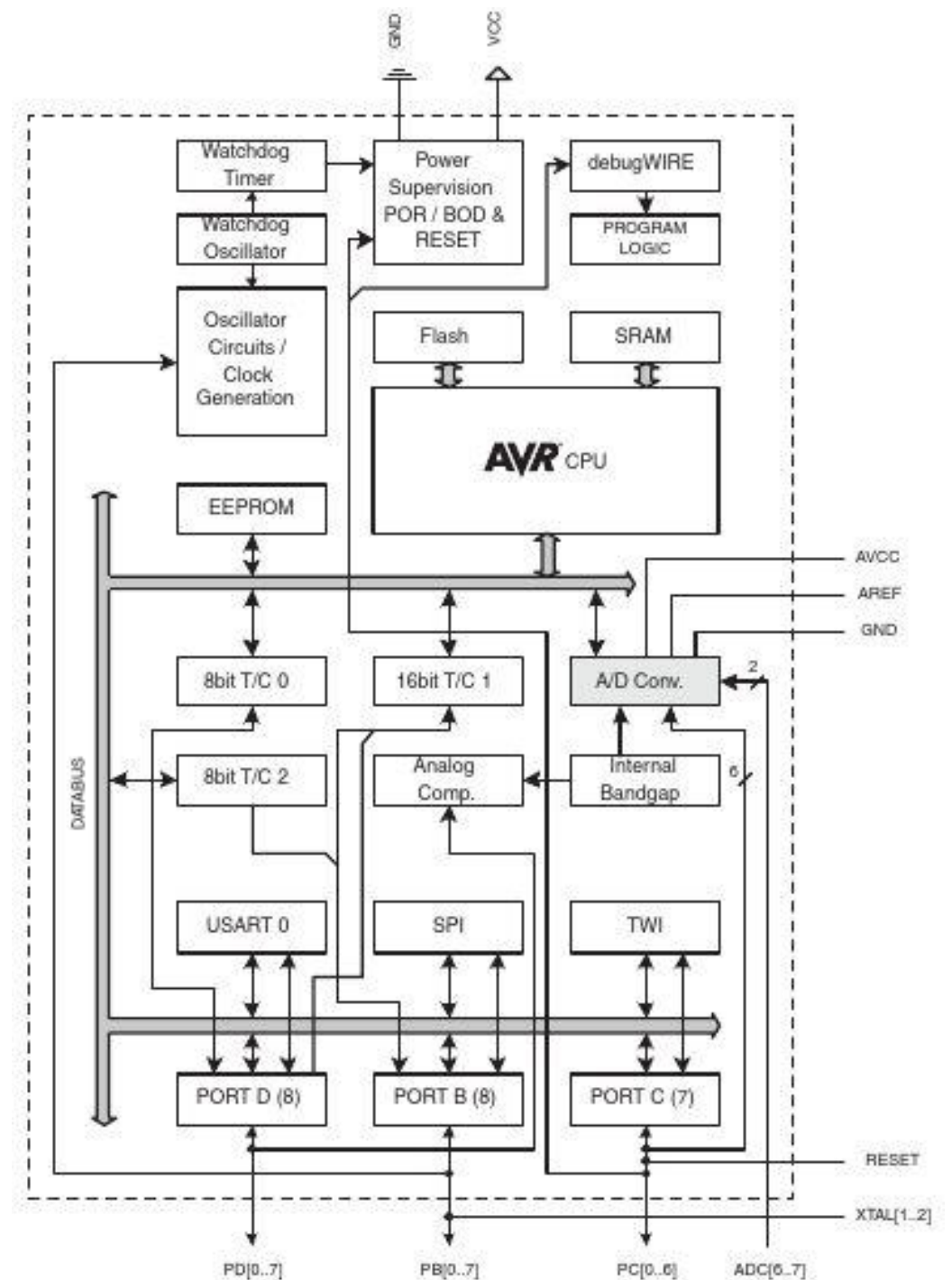
Arduino Pin Mapping



www.arduino.cc

analog input 5
analog input 4
analog input 3
analog input 2
analog input 1
analog input 0

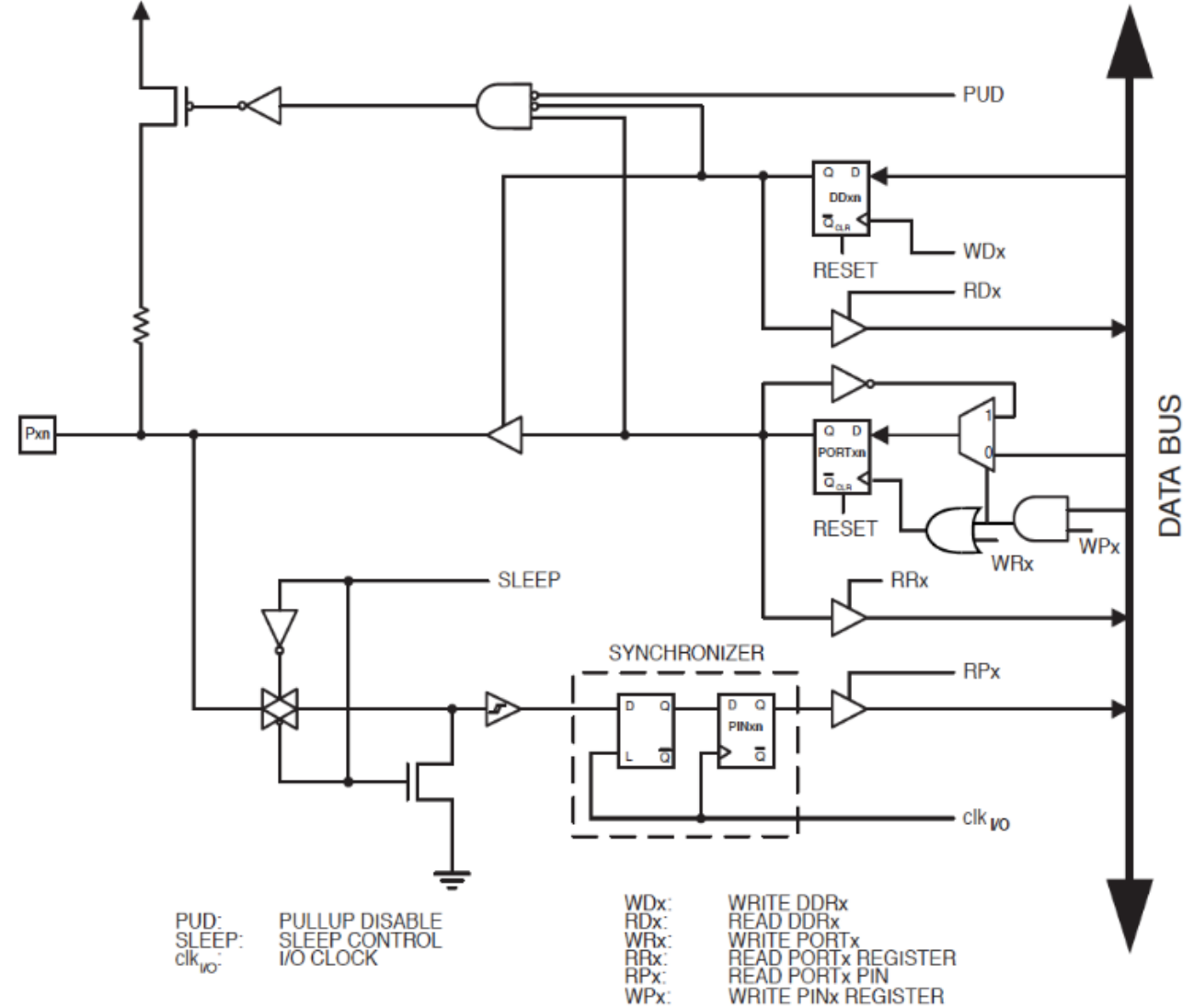
digital pin 13 (LED)
digital pin 12
digital pin 11 (PWM)
digital pin 10 (PWM)
digital pin 9 (PWM)



Grada ulazno/izlazog porta

- 3 8-bitna Porta (B, C, D)
 - Svaki port kontroliran je pomoću tri 8-bitna registra
 - Pojedini bit kontrolira jedan pin
- DDRx
 - Direction register → definira da li je pin ulaz (0) ili izlaz (1)
- PINx
 - Pin Input value → Čitanje s tog registra daje vrijednost postavljenu na pinu
- PORTx
 - Pin output value → upisivanje u ovaj registar postavlja se vrijednost na izlaz

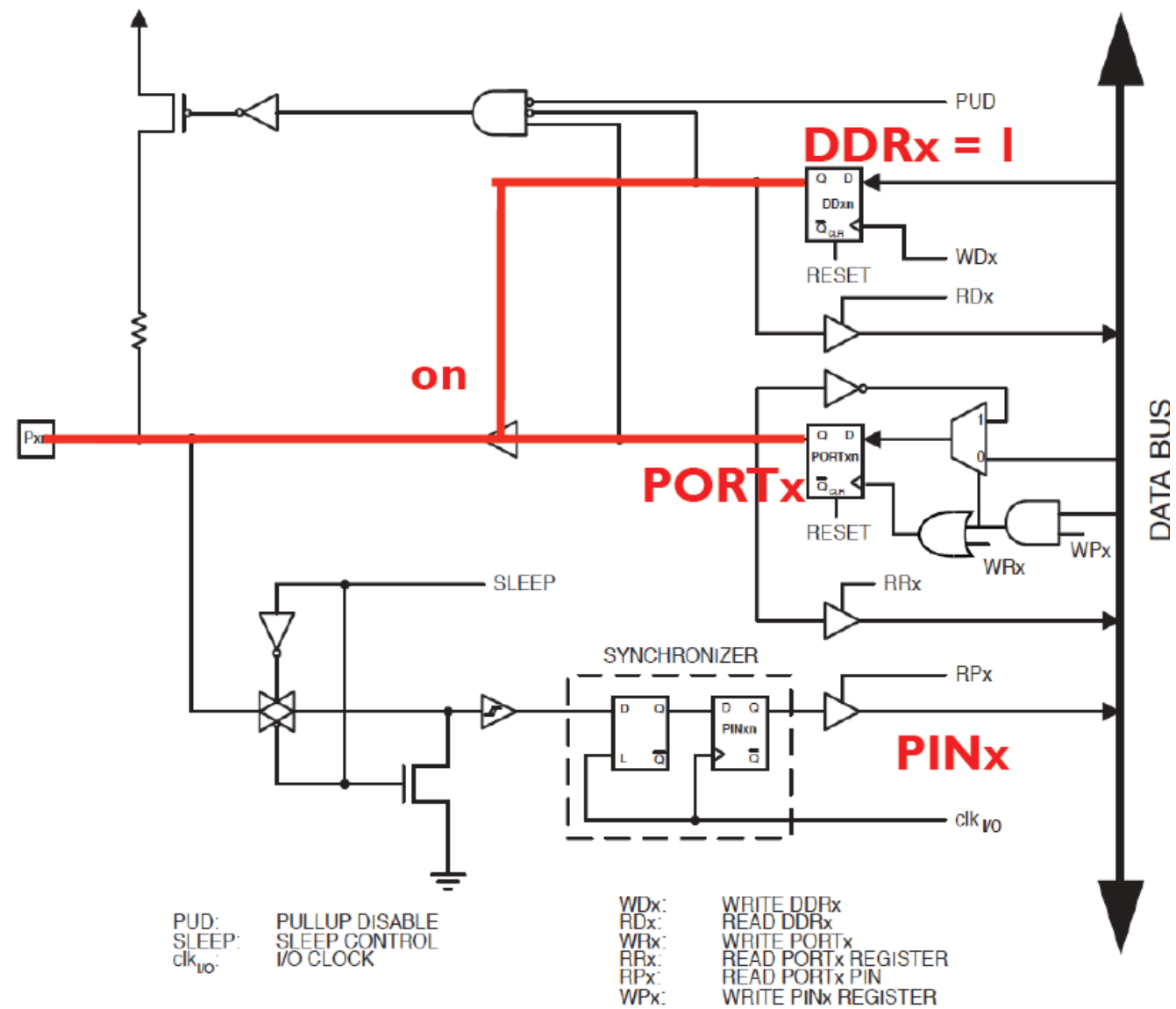
Shema izvedbe digitalnog pina



Digitalni izlazi

- Digitalni izlazi
 - Pinovi 0-13
 - Pinovi 3, 5, 6, 9, 10, 11 → Analogni izlaz → PWM
 - „Analogni izlaz” (ne postoji)
 - Mogućnost postavljanja vrijednosti od 0-255
 - PWM modulacija @ 490 Hz
- Programska podrška
 - `digitalWrite(pin, value);` Arduino IDE

IZLAZ



Podేశavanje registara – primjer PORT B

PORTB – The Port B Data Register

Bit	7	6	5	4	3	2	1	0
0x05 (0x25)	PORTB7	PORTB6	PORTB5	PORTB4	PORTB3	PORTB2	PORTB1	PORTB0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

DDRB – The Port B Data Direction Register

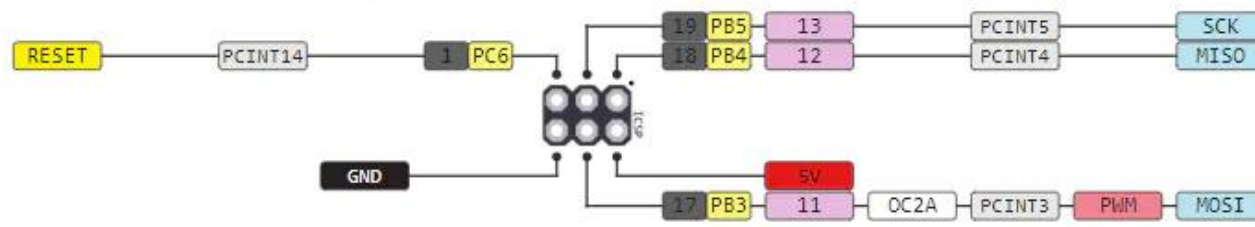
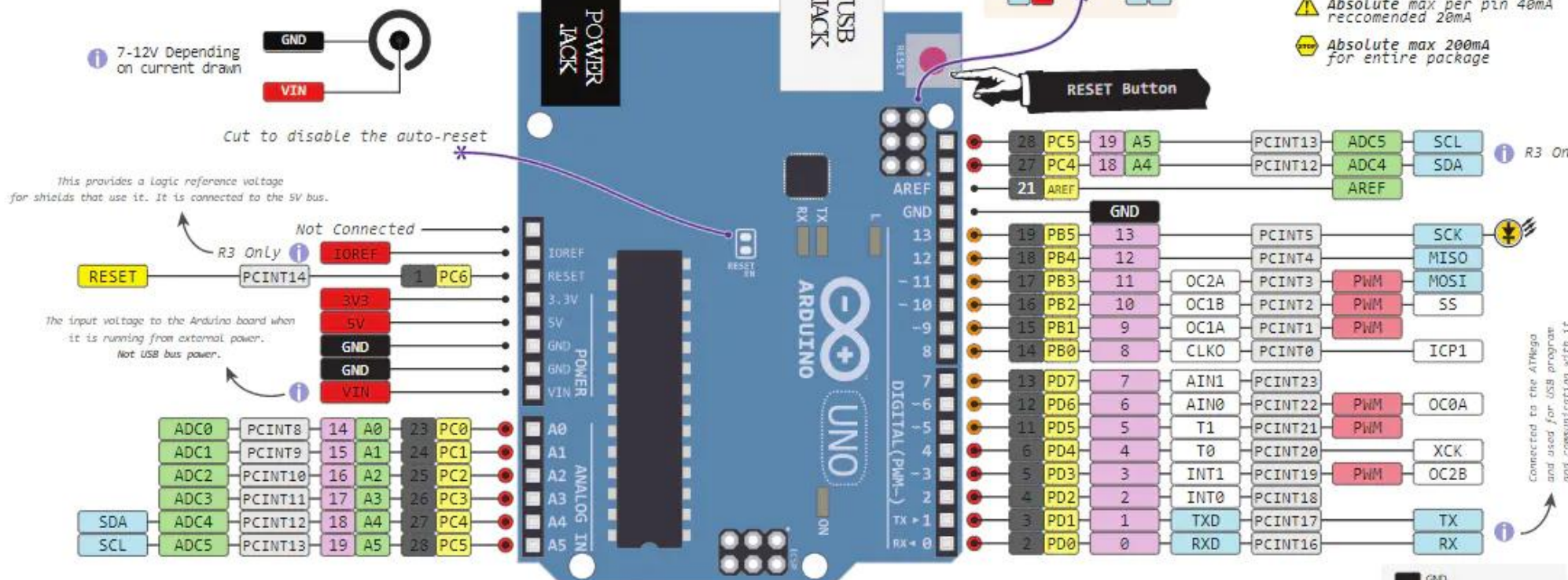
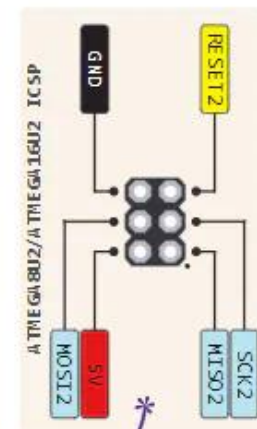
Bit	7	6	5	4	3	2	1	0
0x04 (0x24)	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

PINB – The Port B Input Pins Address

[illegible]

 Absolute max per pin 40mA
reccomended 20mA

 Absolute max 200mA
for entire package



- GND
 Power
 Control
 Physical Pin
 Port Pin
 Pin Function
 Digital Pin
 Analog Related Pin
 PWM Pin
 Serial Pin
 IDE
 Source Total 150mA

Primjer: upravljanje LED žaruljicom na priključku 13 Arduinoa UNO

start:

SBI DDRB,5

blink:

CALL delay

SBI PORTB,5

CALL delay

CBI PORTB,5

JMP blink

delay:

LDI R16, 64;1 strojni ciklus

Delay1:

LDI R17, 125;1 strojni ciklus

Delay2:

LDI R18, 250 ; 1 strojni ciklus

Delay3:

DEC R18 ; 1 strojni ciklus

NOP ; 1 strojni ciklus

BRNE Delay3 ; 2 strojni ciklus

DEC R17 ; 1 strojni ciklus

BRNE Delay2 ; 2 strojni ciklus

DEC R16 ; 1 strojni ciklus

BRNE Delay1 ; 2 strojni ciklus

ret

Zadatak 1.

- Programiraj mikroupravljač koji upravlja (pali-gasi) LED diodom na priključku 6 (Arduino mapirano) na sljedeći način
 - 2.5 sekunde upaljena – 2.5 sekunde ugašena

Zadatak 2.

- Programiraj mikroupravljač koji upravlja (pali-gasi) LED diodom na priključku 4 (Arduino mapirano) na sljedeći način
 - 1 sekundu upaljena – 5 sekundi ugašena

Zadatak 1

- Programiraj mikroupravljač koji upravlja (pali-gasi) LED diodama (priključenih na priključke PB0, PD6, PD7) na sljedeći način:

